

Shourya Baranwal

baranwal.shourya@ucalgary.ca

Linkedin: [linkedin.com/in/shourya-baranwal](https://www.linkedin.com/in/shourya-baranwal)

Cell: (587) 224-5944

EDUCATION

Bachelor of Science, Mechanical Engineering

May 2029

Schulich School of Engineering, University of Calgary

Dean's List Recipient - Fall 2024, Winter 2025

1st place in GHacks 2025 - Hexagon

Available for Internship January 2027-December 2027

SKILLS

Technical Skills: SolidWorks, Onshape, Altium, Computer Literacy, Excel, Arduino, Robotics, Motor Learning

Interpersonal Skills: Team Coordination, Management, Leadership, Critical Thinking

Programming Languages: Python, Java, C++, MATLAB, Simulink, State flow

WORK EXPERIENCE

Engineering Manager

September 2024 - Present

University of Calgary Solar Car Team, Calgary AB

- Act as technical link between mechanical, electrical, and software sub teams overseeing 12 sub teams overall
- Facilitated the shop access and equipment training for over 20+ in collaboration with the Engineering Makerspace of the University of Calgary
- Race Team 2026: Partnered with suspension leads to assemble double wishbone suspension system to get the car rolling
- Connected with over 15 teams had experience building Solar Cars including Georgia Tech, University of Toronto, McMaster University
- Partook in preparing for and attending a 11 stationed rigorous scrutineering process for solar car
- Implemented a gantt chart outlining clearly communicated key projects and deadlines across 12 engineering sub-teams

Researcher

May 2026 - Present

Integrative Sensorimotor Neuroscience Laboratory, Calgary, Canada

- Modelled and programmed reward algorithm for force-field reaching tasks to automatically compare current trial performance with 5 individual past trials
- Extracted and analyzed reaching data for 10+ participants led to 36% more accurate reaching movements
- Creating a task in Simulink and State flow representing balancing an inverted pendulum with the KinArm to enhance motor learning

PROJECTS

Door Hinges, Solar Car

March 2025 - Present

- Took initiative to model design in Solidworks for lift-up opening mechanism for solar car door
- Simulated a predicted static load of 10 kg on the door hinge leveraging ANSYS, resulting in high factors of safety and low stress concentration points around the hinge

Junior Achievement Associate

November 2023 - March
2025

- Promoted to VP of Finance due to reliability and proven dedication
- Analyzed consumer patterns and led company to 3 times the profit relative to initial investments

First Tech Challenge

January 2023 - April 2024

- Secured 1st place in Alberta's qualifier cup and Think Award
- Designed the robot using Onshape for easier collaborative work across a team of 20 members and use of iterative design to master endgame tasks